

Vaibhav Suresh Parekh

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EDUCATION

Carnegie Mellon University

Master of Science – Robotics, Mechanical Engineering (Research) | GPA: 3.84/4.00

Pittsburgh, PA

May 2026

University of Mumbai (K. J. Somaiya College of Engineering)

Bachelor of Technology – Mechanical Engineering | GPA: 9.14/10.00

Mumbai, India

May 2022

SKILLS

Tools: Linux, Ubuntu, ROS, ROS2, Gazebo, Nvidia Isaac Sim, Webots, RViz, Foxglove, Git, Docker, SolidWorks, ANSYS

Languages & libraries: C++, Python, MATLAB, Simulink, CMake, OpenCV, MediaPipe, YOLO, PyTorch, CUDA

WORK EXPERIENCE

Lightspeed Robotics

Northbrook, IL

Software and Robotics Engineering Intern

Sep 2025 - Dec 2025

- Built an **LLM-based parser** for BTLx (XML schema for timber fabrication) to translate design data into robotic panel-cutting instructions, enabling AI-driven construction automation.
- Developed simulation of an articulated mobile manipulator robot in **NVIDIA Isaac Sim** integrated with **Movelt** (manipulation) and **Nav2** (navigation) in ROS2, incorporating vision modules for scene understanding.

Inductive Robotics

Austin, TX

Robotics Engineer Intern

Jul 2025 - Aug 2025

- Authored URDF, SDF and built a **ROS2**-integrated USD simulation environment in **Blender** to enable autonomous docking simulation of a Clearpath Jackal mobile robot with an inductive charging station in **NVIDIA Isaac Sim**.
- Generated **synthetic dataset** in Isaac Sim and trained **YOLOv11** model for detecting vehicles reversing through taillight-state classification, achieving **92% accuracy** on real-world data through effective sim-to-real using domain randomization.

Tata Power Renewable Energy Limited

Mumbai, India

Lead Associate

Jun 2022 - Aug 2024

- Performed financial and energy modeling using Plexos and Excel to integrate over **6 GW** of energy from wind, solar, and ESS for optimizing power plant capacities to meet diverse power requirements.
- Led a cross-functional project to customize and **fine-tune a GPT-3.5 model** for summarizing, assessing risks, and comparing lengthy tender documents, **reducing TAT** for tender analysis from about 45 minutes to less than 10 minutes.

ACADEMIC PROJECTS

Graduate Research (Computational Engineering & Robotics Lab)

Oct 2024 - May 2026

- Devised an unsupervised approach to identify keypoints on multi-viewpoint objects from single RGB image using stacked Hourglass CNNs, achieving **PCK@10 accuracy of 82.8% and 69.5%** for intra- and cross-instance correspondences.
- Rigged robot models in Blender for simulation in Nvidia Isaac Sim, enabling visual **SLAM** implementations and testing.

F1TENTH Autonomous Racing

Jan 2026 - May 2026

- Engineered a **LiDAR-based ROS2 navigation stack** (PID Wall Following, Follow-the-Gap, Pure Pursuit, RRT*, MPC) deployed on **Nvidia Jetson** for an autonomous racecar, utilizing SLAM Toolbox for **mapping** and Particle Filter for **localization**.
- Achieved 1st place among 9 teams with a **44.6% lower lap time** than the overall average through **visual odometry/VESC** tuning.
- Built an online anti-skid system using **sensor fusion** of wheel-encoder and **IMU** velocities to detect slippage and adaptively reduce waypoint speed limits.

Neural 3D Reconstruction and Generative Modeling

Oct 2025 - Nov 2025

- Built **diffusion**-guided optimization pipelines in **PyTorch3D** using Score Distillation Sampling (SDS) and **3D Gaussian Splatting**, applied to image, mesh, and **NeRF** based **text-to-3D** generation tasks.
- Implemented Neural Radiance Fields (NeRF) for novel view synthesis, **point cloud processing**, single-view 3D prediction, and differentiable rendering.

Self-Docking Autonomous Robot

Jan 2025 - May 2025

- Led the perception stack to develop a wall-outlet detection pipeline using **YOLOv8** with custom dataset on **Intel RealSense D435**, deployed on **Raspberry Pi 5** and integrated via **ROS2** for visual SLAM via RTAB-MAP and planning via Nav2.

Planning for Multi-UAV Search and Rescue

Oct 2024 - Dec 2024

- Formulated and simulated an **NBV (next-best-view) planner** algorithm in **MATLAB** leveraging LiDAR-based occupancy maps to identify intermediate goal positions, maximizing UAV coverage efficiency in disaster zones.
- Deployed **RRT algorithm** for efficient UAV navigation to intermediate goal points, enabling faster search-and-rescue coverage.

Controller Design for an Autonomous Buggy

Oct 2024 - Dec 2024

- Programed **PID**, **LQR**, and **Kalman** filter-based controllers for a car to accurately follow predefined trajectories in **Webots**.
- Implemented **A*** for path planning, and **EKF SLAM** for localization and controlling the vehicle in GPS-denied environment.